

SISIMIUT HOLSTEINSBORG, Greenland

WMO: 042340

Lat: 66.9510N

Lon: 53.7200W

Elev: 8

StdP: 101.23

Time Zone: -3.00 (GRL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-26.5	-23.8	-31.2	0.2	-25.0	-28.8	0.3	-22.3	17.2	1.9	14.8	-1.5	1.8	100	0.882

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.4	16.4	9.5	14.2	8.9	12.2	8.1	10.5	14.4	9.5	13.0	8.6	11.6	6.2	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.7	7.0	10.6	7.5	6.4	10.0	6.6	6.1	9.2	30.3	14.6	28.2	13.1	26.3	11.8	14.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.7	10.8	9.3	DB	-26.2	19.8	4.2	2.6	-29.3	21.6	-31.7	23.1	-34.1	24.6	-37.2	26.4	(4)
(5)			WB	-26.6	12.2	4.0	1.6	-29.5	13.3	-31.8	14.2	-34.1	15.1	-37.0	16.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.2	-11.1	-12.6	-11.7	-4.9	0.8	5.5	8.1	7.6	4.1	-0.8	-4.6	-7.8	(6)	
	DBStd	8.57	6.17	7.10	6.81	4.91	3.42	3.04	3.02	2.56	2.38	3.39	4.22	4.82	(7)	
	HDD10.0	4491	653	632	672	448	286	139	75	82	177	337	438	552	(8)	
	HDD18.3	7504	912	866	930	698	544	386	318	332	427	594	688	810	(9)	
	CDD10.0	28	0	0	0	0	0	3	16	9	1	0	0	0	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	3.5	4.1	3.4	3.3	3.4	3.3	3.4	3.2	3.1	3.4	3.3	3.9	4.0	(14)	
(15) Precipitation	PrecAvg	590	57	36	35	36	32	36	47	65	66	56	64	61	(15)	
	PrecMax	822	157	86	92	81	58	75	91	109	142	117	134	128	(16)	
	PrecMin	392	5	14	3	13	8	12	18	23	19	16	24	26	(17)	
	PrecStd	106	35	17	21	18	14	15	21	22	26	22	27	25	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.2	6.8	6.2	7.1	13.1	17.7	20.1	18.0	12.4	9.7	8.3	5.9	(19)	
		MCWB	1.4	1.4	2.2	3.4	6.8	9.4	11.1	10.6	7.6	4.4	4.1	1.5	(20)	
	2%	DB	2.8	3.4	2.9	4.4	8.9	14.5	17.2	15.2	10.0	6.7	5.0	3.2	(21)	
		MCWB	-0.1	0.1	0.4	1.5	5.0	8.0	10.0	9.7	6.0	2.9	1.7	0.9	(22)	
	5%	DB	0.0	0.3	0.6	2.8	6.9	11.9	14.7	13.0	8.5	4.9	2.9	1.1	(23)	
		MCWB	-2.0	-1.9	-1.4	0.7	3.6	7.1	9.1	8.9	5.2	2.3	0.7	-0.9	(24)	
	10%	DB	-2.5	-3.0	-1.9	1.5	5.1	10.0	12.7	11.2	7.4	3.4	1.3	-1.2	(25)	
		MCWB	-4.3	-4.6	-3.5	-0.2	2.4	6.3	8.5	8.0	4.7	1.2	-0.4	-3.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.8	2.2	2.8	3.8	7.2	10.2	12.2	11.7	9.1	5.5	5.2	2.7	(27)	
		MCDB	4.8	5.2	5.8	6.5	11.4	15.8	17.1	16.1	11.0	7.7	7.7	4.6	(28)	
	2%	WB	0.0	0.2	0.5	2.0	5.4	8.6	10.7	10.4	6.7	3.7	2.4	1.0	(29)	
		MCDB	1.8	3.4	2.5	3.6	8.5	13.1	15.1	13.6	8.8	5.8	4.0	3.1	(30)	
	5%	WB	-2.0	-1.9	-1.3	1.0	3.9	7.5	9.7	9.5	5.9	2.5	0.9	-1.0	(31)	
		MCDB	0.1	0.6	0.6	2.4	6.2	11.2	13.7	12.3	7.7	4.4	2.6	0.9	(32)	
	10%	WB	-4.2	-4.6	-3.6	-0.1	2.8	6.5	8.8	8.6	5.1	1.5	-0.4	-2.9	(33)	
		MCDB	-2.5	-2.9	-1.8	1.5	4.6	9.6	12.2	10.9	6.9	3.2	1.3	-1.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.0	5.5	5.9	5.6	5.1	6.2	6.4	5.2	4.2	4.1	4.4	4.5	(35)	
		MCDBR	7.9	9.0	7.7	6.5	8.6	10.8	11.0	9.0	6.0	5.6	6.4	6.2	(36)	
	5% WB	MCWBR	6.2	7.0	5.8	4.6	5.1	5.5	5.1	4.5	3.6	4.0	5.1	4.7	(37)	
		MCWBR	7.7	9.0	7.7	5.9	8.1	9.8	9.7	8.0	5.1	4.9	6.0	6.1	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.170	0.222	0.255	0.313	0.320	0.332	0.333	0.334	0.293	0.248	0.356	N/A	(40)	
		taud	1.929	2.213	2.316	2.239	2.381	2.476	2.520	2.528	2.540	2.260	2.850	N/A	(41)	
	Edn at Noon	Edn at Noon	219	657	823	842	880	877	865	825	783	596	243	0	(42)	
		Edh at Noon	13	45	71	101	99	93	86	77	59	42	16	0	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.05	0.51	2.00	3.92	5.40	5.83	5.28	3.64	2.10	0.77	0.12	0.00	(44)	
	RadStd	RadStd	0.01	0.06	0.17	0.34	0.32	0.52	0.51	0.28	0.21	0.07	0.01	0.00	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		+1.19	N/A	N/A	+2.14	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+0.91	N/A	N/A	+0.82	N/A	N/A	N/A	N/A	+8	N/A

Nomenclature: See separate page